access will be denied. Assuming critical resources have been correctly categorized, this policy requires people to use properly functioning managed devices when accessing them.

1.4.2 Policy as Data: Storing and managing access rules dynamically

In contrast to Policy as Code, which focuses on general rules that can be evaluated to determine access, Policy as Data uses a store of information about specific attributes, permissions, or relationships that a general-purpose engine can use to determine access at runtime.

The data is structured and usually stored in some kind of database. End-users, administrators, or business systems define the rules dynamically using a user interface or an API. For example, when a customer chooses to share a Google document with members of their same organization, that permission is stored in the relationship the customer has with their organization. Figure 1.3 shows the relationship graph for a Google document. This could be stored in a graph database for use by the authorization engine. While this may seem like an ACL, it’s much richer, and the additional information allows general-purpose relationships to be expressed.